Master Product Requirements Document (PRD v4.0)

Product Name: EDITH Next-Gen AML Intelligence Platform (formerly iShield)
Target Architecture: Modular Microservices (.NET 10 LTS + React 19 Portal + OpenSearch 2.12 + PostgreSQL 17)
Target Market: GCC Financial Institutions, Exchange Houses, and Commercial Banks
Screening Scope: English & Transliterated Phonetic Optimized
Performance SLA: Sub-second 5-pass screening budget (P99 < 10ms in-memory / < 50ms OpenSearch)
Database Footprint: Dual-Database (PostgreSQL 17 + OpenSearch 2.12 — Zero Redis Dependency)
Version: 4.0
Last Updated: October 2026

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1. Executive Summary & Strategic Scope

The **EDITH AML Intelligence Platform** modernizes legacy monolithic compliance software into a cloud-native, modular microservices platform running on **.NET 10 LTS** and **React 19**. The platform is engineered to eliminate alert fatigue, accelerate compliance investigations, and automate regulatory reporting for financial institutions operating under FATF-aligned central bank mandates (CBUAE, SAMA, Central Bank of Oman).

Core Business Objectives

- **Plug-and-Play Modularity:** Decouple platform capabilities into isolated core modules and optional add-ons, enabling feature activation and licensing tiers without modifying the underlying codebase.
- **True Asynchronous Microservices:** Heavy background processes (bulk screening, reporting, delta syncs) execute asynchronously via `HTTP 202 Accepted` and SignalR notifications while users continue navigating the platform.
- **Dual-Mode Core Banking System (CBS) API Integration:** Synchronous pre-clearance (`CLEARED` / `HELD`) paired with asynchronous HMAC-SHA256 signed webhook callbacks (`RELEASE` / `FREEZE`) when compliance officers adjudicate held transactions.
- **Streamlined Dual-Database Footprint:** Zero Redis dependency. Relies on PostgreSQL 17 for relational ACID integrity and OpenSearch 2.12 for phonetic/fuzzy entity resolution.
- **Live React 19 Operator Workstation:** Unified 14-view compliance portal (`edith-portal`) bridged to `Edith.Gateway` with real JWT role switching (`ANALYST` , `MLRO` , `SUPER_ADMIN` , `AUDITOR`).

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2. Modular Architecture & Core System Integration

2.1. Core Banking System (CBS) API Integration Strategy

- **Synchronous Pre-Clearance:** The Core Banking System sends `POST /api/v1/screen/transaction` with an `Idempotency-Key` header. EDITH executes the 5-pass engine, auto-provisions an investigation case in `cases.aml_cases` if `HELD` , and returns immediately.

- **Asynchronous Signed Webhook Callbacks:** When an MLRO resolves a held case as `ClearedFalsePositive` (`RELEASE`) or `ReportedStrFreeze` (`FREEZE`), `ICbsWebhookDispatcher` dispatches an HMAC-SHA256 signed `AmlDecisionWebhookPayload` (`X-Edith-Signature`) back to the CBS webhook endpoint.

2.2. Schema-per-Module Isolation (PostgreSQL 17)

Every module manages its own isolated PostgreSQL schema (`core`, `screening`, `cases`, `rules`, `notifications`).

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3. Dual-Database Persistence Architecture (No Redis)

Database / Store	Technology	Scope & Responsibility
System of Record & Audit	**PostgreSQL 17**	Master entity catalog, case workflows, customer KYC records, audit trails, user IAM, custom rule configurations, bulk job metadata, skipped duplicate records, and whitelist tables.
Search & Screening Engine	**OpenSearch 2.12**	Read-optimized search index (`edith_watchlist_v1`, `edith_customers_v1`) with custom n-gram and Double Metaphone phonetic token analyzers, plus automatic in-memory fallback when offline.

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4. Multi-Pass Screening Engine & Bulk Ingestion Pipeline

4.1. The 5-Pass Real-Time Screening Lifecycle

1. **Pass 1 — Normalization & Cleansing:** ASCII folding, punctuation removal, case folding, corporate suffix stripping.
2. **Pass 2 — Broad Retrieval:** OpenSearch DisMax query + phonetic token matching pulls top candidate matches.
3. **Pass 3 — Phonetic & String Distance Scoring:** Double Metaphone, Jaro-Winkler, and Damerau-Levenshtein evaluation.
4. **Pass 4 — Multi-Entity Evaluation:** Simultaneously screens Sender, Beneficiary, and Intermediary/Correspondent Bank.
5. **Pass 5 — Deterministic Scoring & Whitelist Bypass:** Calculates weighted composite score; suppresses hits if active in PostgreSQL `screening.whitelists`; auto-creates an `Open` case if score $\ge 80%$.

4.2. Streaming Bulk Ingestion & Skipped Duplicate Export (`MOD-SCRN-BULK`)

- **Supported Formats:** Streaming parser (`IAmlStreamingParser`) supports **CSV**, **XML**, and **JSON** mapped to 9 standardized fields (`AML_TYPE`, `AML_FULLNAME`, `AML_IDDETAILS`, `AML_CATEGORY`, `AML_DATEOFBIRTH`, `AML_REMARKS`, `AML_NATIONALITY`, `AML_PROGRAM`, `AML_ALIASNAME`).
- **Duplicate Detection & Keep-Skipped Option:** Entities already present in PostgreSQL/OpenSearch are automatically skipped during bulk upload. When `keepSkipped=true`, skipped entities are retained and can

be downloaded on demand in the user's chosen format (`CSV` , `JSON` , or `XML`) via `GET /api/v1/bulk/jobs/{id}/skipped?format=csv|json|xml` .

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5. Case Investigation Engine & 4-Eyes Adjudication (`MOD-CASE-WORK`)

- **Pessimistic 15-Minute Lease Locking:** Only one analyst can hold the active investigation lock on a case at a time (`POST /api/v1/cases/{id}/acquire-lock`). Competing lock attempts return `409 Conflict` .
- **Multi-Type Investigation Notes:** Supports `INVESTIGATION` , `SUPERVISOR` , and `RATIONALE` notes persisted in real time and broadcast via SignalR `CaseHub` .
- **4-Eyes Maker-Checker Governance:**
  - **Analyst (Maker):** Investigates token breakdowns, adds notes, and escalates cases to `PendingApproval` (`Escalate Case to MLRO Queue`). Cannot execute final `Resolved` disposition.
  - **MLRO / Compliance Officer (Checker):** Reviews escalated cases and executes final adjudication (`ClearedFalsePositive` $\rightarrow$ CBS `RELEASE` , or `ReportedStrFreeze` $\rightarrow$ CBS `FREEZE` + UNODC `goAML` STR XML generation).

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6. Master Feature Matrix & Packaging

Module Code	Module / Feature Name	Package Tier	Modernized Implementation
`MOD-CORE-IAM`	Multi-Tenant IAM & RBAC	**Core**	JWT Bearer auth, dynamic role persona switching, tenant header isolation.
`MOD-CORE-API`	Core System Gateway (`Edith.Gateway`)	**Core**	Dual-mode API (Sync pre-clearance + Async HMAC-SHA256 CBS Webhooks) with Idempotency.
`MOD-SCRN-BASE`	Watchlist & Sanctions Screening	**Core**	5-pass OpenSearch 2.12 + In-Memory hybrid engine, Double Metaphone, Jaro-Winkler.
`MOD-SCRN-BULK`	Bulk File Screening Engine	**Core**	Streaming CSV/XML/JSON ingestion, duplicate skipping, and multi-format skipped export.
`MOD-RULE-PROC`	AML Transaction Rule Engine	**Core**	Real-time corridor/threshold/structuring rules and historical backtesting sandbox.
`MOD-CASE-WORK`	3-Pane Case Investigation Canvas	**Core**	15-min pessimistic locking, 4-Eyes Maker-Checker workflow, and CBS webhook dispatch.
`MOD-GOAML-GCC`	Automated `goAML` Exporter	**Add-on**	UNODC goAML v5.0 XML payload generator for GCC Financial Intelligence Units.
`MOD-AI-NARR`	GenAI STR Narrative Copilot	**Add-on**	Automated 5-section regulatory STR narrative drafting from case telemetry.
`MOD-AI-GRAPH`	Entity Relationship Visualizer	**Add-on**	Multi-hop network graph builder detecting UBO and circular layering rings.